BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Autumn 2 Assessment

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What protection does encryption provide if stored data is stolen?

2. What is POP used for?

A) Sending new email
B) Transferring web pages
C) Downloading email
D) Looking up IP addresses

3. What is FTP used for?

4. A network component changes but keeps its interface. "Changing one protocol layer means redesigning every layer." Correct this. Why are layers useful?

Today's objectives

Complete a timed assessment independently and show all working. Use OCR command words and mark allocations accurately.

Key vocabulary - Read, cover, recall

State
Give a correct fact with little or no elaboration.

Describe
Give characteristics or relevant details.

Explain
Give a reason and link it to an outcome.

Compare
Identify relevant similarities or differences.

Write one key term and its meaning in your own words:

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Learn and practise

New learning - Key ideas

Assessment conditions	This lesson uses a 45-minute independent assessment. Follow your teacher's instructions on access to notes, timing and permitted resources.
Read the command word	State = brief fact. Describe = give relevant detail. Explain = link the cause/feature to its effect. Compare = show a relevant similarity or difference.
Plan extended answers	Use the number of marks to decide how many distinct points and explanations are needed. Write clearly and directly.
Work independently	Do not reveal model answers during the timed section. After the assessment, use the separate answer-review resource when your teacher allows it.

I Do - One worked example

A familiar practice question asks: explain why a larger cache can improve CPU performance. [2 marks] How would you plan TWO linked points? This is not an assessment question.

1. The command word is 'explain', so a definition alone is insufficient.
2. A larger cache holds more frequently used data/instructions [1].
3. This reduces the need for slower access to RAM [1].
4. Two linked ideas gain two marks. Do not write this model for an unrelated question.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. Practice only: a single school building needs to share a printer. Give TWO reasons to use a LAN. [2 marks]

2. Practice only: a school receives a fake password-reset email. State the likely attack and one warning sign. [2 marks]

3. Practice only: explain one benefit of adding CPU cores when running several applications. [2 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Assessment coding question: independently predict, trace and change the supplied OCR ERL program. Do not run code until instructed.

```
accessLevel = 1
if accessLevel > 1 then
print("Staff access")
else
print("Standard access")
endif
```

BLACK PEN: predict or complete the code; test, then improve.

You Do - Independent exam questions

Q2. Assessment Q1. State what LAN and WAN mean, and identify the device that forwards data between networks. [3 marks]

Q3. Assessment Q2. A public library uses a school-style star LAN to share files and printers. Explain one advantage of the topology and one potential weakness. [4 marks]

Q4. Assessment Q3. A student is sent a fake link asking for school credentials. Explain the threat and TWO protective measures. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. For a four-mark explain question, listing one unsupported fact always earns full marks. ☐ **T** ☐ **F**
2. This lesson uses a 45-minute independent assessment ☐ **T** ☐ **F**
3. State = brief fact ☐ **T** ☐ **F**

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Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Without the key, an attacker cannot easily read ciphertext. Encryption does not stop theft or deletion. ☐

Q2 C - Downloading email ☐

Q3 Transferring files between networked computers. ☐

Q4 Layers split communication into tasks and provide services to each other. Clear interfaces let one layer change without changing all the others. This helps compatibility, development and finding faults. ☐

Code First - Model

With accessLevel = 1: Standard access [1]. With 2: Staff access [1].

Possible extension:

```
if accessLevel > 2 then
print("Extra check")
endif
```

Award [1] for correct IF condition and [1] for printing the specified message with correct OCR ERL syntax. Model scheme for original practice.

We Do - Check the model

1. LAN covers a small geographical area [1]; its devices can share a network printer [1]. ☐

2. Phishing [1]; suspicious link, urgency or unexpected request for credentials [1]. ☐

3. Different cores may process more tasks simultaneously [1], improving multitasking [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Local Area Network [1]; Wide Area Network [1]; router [1]. ☐

Q3 Central switch simplifies adding/managing devices [1], allowing easier administration [1]. The central switch is a single point of failure [1], so if it fails connected devices may be unable to communicate [1]. ☐

Q4 Phishing/social engineering [1] tricks users with a false link or identity [1]. Training helps recognise/report suspicious messages [1], reducing credential disclosure [1]. Two-factor authentication adds separate proof [1], reducing risk when a password is stolen [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: